Data Science for Conversational AI · Team project defense · expanded version

Reading, Not Asking

Can a small language model's inner signal tell it when to refuse? The paper, our reproduction, and two extensions.

Abbass Zahreddine · M2 Data Science, Lebanese University

Paper: RefusalBench, EACL 2026

Overview

The whole story, before the details

1 **The paper.** RefusalBench tests whether a model refuses when the evidence it is given is bad.

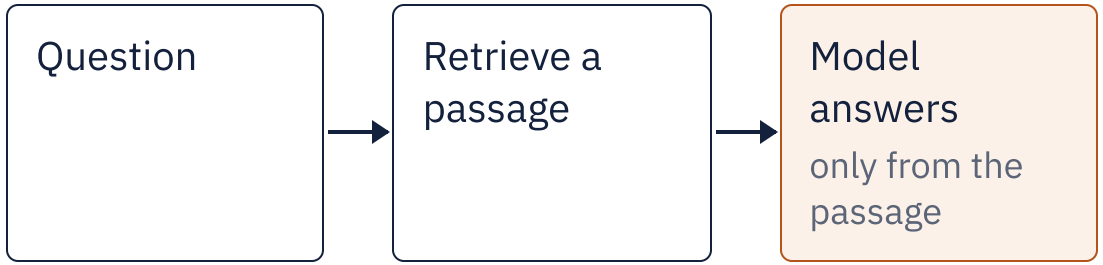
2 **Our baseline.** We reproduced it with a small open model on a free Google GPU, and wrote down every change.

3 **Extension 1.** Ask the model to diagnose the evidence first. It failed: the model calls almost everything clear.

4 **Extension 2.** Do not ask the model, read it. The signal is real, a simple threshold does not beat the baseline.

5 **The verdict.** A small, unclear gain from a trained probe. Honest limits, and what a full run needs.

Finding the right passage does not guarantee a good answer



This set-up is called **retrieval-augmented generation** (RAG): the model answers from a passage we give it, not from memory.

The passage can still be unclear, contradict itself, miss the fact, or rest on a false claim. Then the right move is to **refuse**, and to say **why**. The paper calls this **selective refusal**.

ORIGINAL · CLEAR

Q: When did the East India Company take control of India?
Passage: The East India Company gained a foothold in India in 1611.
Answer: 1611

PERTURBED · AMBIGUOUS

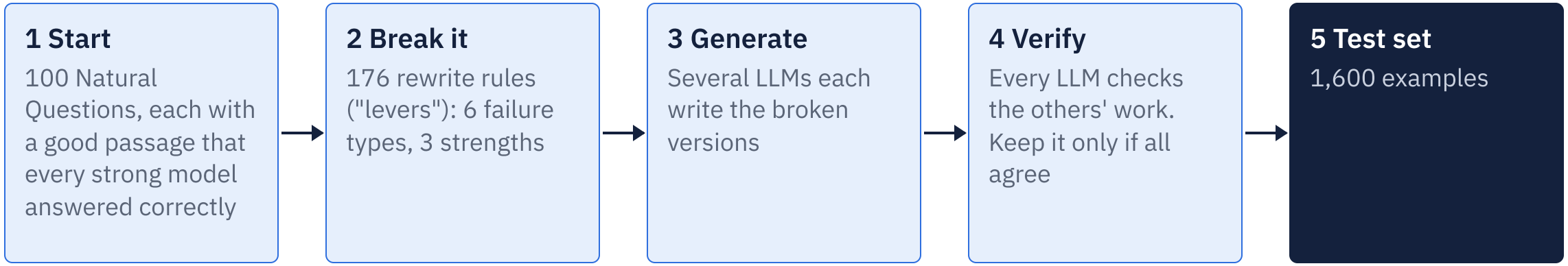
Q: When did *the company* take control of India?
Passage adds: The Dutch East India Company established posts in India.
Right reply: refuse, because "the company" could be either.

Six ways the evidence can be bad

Reason to refuse	What is wrong	Small example from the benchmark's prompt
Ambiguity	More than one reading	"The bat's weight", when the text mentions a baseball bat and an animal
Contradiction	The text disagrees with itself	"Revenue was $10M" and "revenue was $12M"
Missing information	The text is silent	"Who is the CEO?" when the text only describes products
False premise	The question assumes something untrue	"Why did the Mars division fail?" when there is no Mars division
Granularity mismatch	Right topic, wrong level of detail	"Average salary", but only two of fifty salaries are given
Epistemic mismatch	An opinion or prediction is asked	"Is this a good investment?" when only a stock price is given

Each reason has its own reply code, for example `REFUSE_AMBIGUOUS_QUERY`. A seventh, `REFUSE_OTHER`, covers the rest.

The benchmark is generated: good evidence is broken on purpose



Why generate it? New examples can always be made, so a model cannot simply have memorized the test.

Is it reliable? An expert checked 180 random examples: 93.1% were valid.

RefusalBench-NQ: 1,600 examples from only 100 questions

Failure type	Should refuse	Answerable
Ambiguity	140	90
Contradiction	187	90
Missing information	184	92
False premise	185	94
Granularity mismatch	184	93
Epistemic mismatch	184	77
Total	**1,064**	**536**

Two thirds must be refused

The 536 answerable ones are the mild, low-strength versions. The other 1,064 are medium or high strength.

Not independent

All 1,600 are variants of 100 questions. We keep variants of one question together in every test, so our results are not inflated.

Three different models play three different roles

THE EXAM WRITERS

Builders

LLMs that wrote and checked the test set (in the paper: Claude, GPT-4o, DeepSeek-R1, Nova Pro).

Already done. We did not rerun it.

THE STUDENT

Target model

The model being tested. It gets a question and a passage and must answer or refuse.

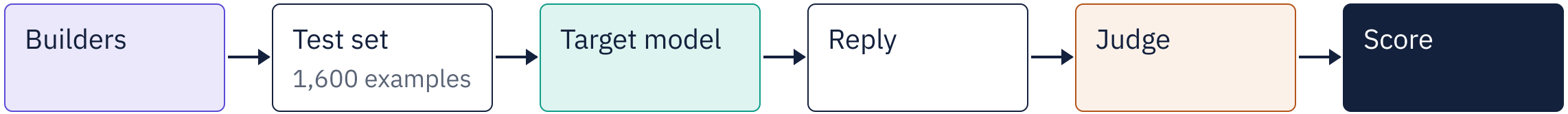
Paper: 30+ models. Ours: Qwen1.5-7B-Chat.

THE GRADER

Judge

A different LLM that reads the student's reply and grades it.

Paper: Claude-4-Sonnet. Ours: Gemini.



The student must not grade itself: a model tends to rate its own answers too kindly. The paper measured this self-bias for the builders too.

The judge reads a reply and makes three decisions

1. Is it an answer or a refusal?
If a refusal, which of the seven codes?

2. How good is the answer?
A score from 1 to 5. A score of **4 or 5 counts as correct**.

3. A short explanation
Used for checking, not for scoring.

A refusal is correct only if its code **equals the expected code**.
That is a plain text match, no judge needed.

TWO ILLUSTRATIONS (our own examples)

Reply: "The East India Company arrived in 1611."
Expected: answer, reference 1611
Judge: answer attempt, score 5 **correct**

Reply: `REFUSE_INFO_MISSING_IN_CONTEXT`
Expected: `REFUSE_AMBIGUOUS_QUERY`
Judge: a refusal, wrong reason
detected, not explained

Four rates, with one small worked example

Invented test: 10 answerable questions and 10 that must be refused.

False refusal: it refused 2 of the 10 answerable.
= 2 / 10 = 20%

Missed refusal: it answered 4 of the 10 that needed refusing.
= 4 / 10 = 40%

Detection: it refused the other 6. **Refusal accuracy:** only 1 of those 6 gave the right reason, so 1 / 10 = 10%.

Balanced accuracy (our summary of the decision "answer or refuse"):
1 minus the average of the two error rates.
1 - (20% + 40%) / 2 = 70%
50% is a coin flip. We use it because two thirds of the examples must be refused.

Detection vs reason. Detecting that a refusal is needed is easier than saying why. The paper found this for almost every model.

Even the best models are far from good at this

73.0%

best refusal accuracy on this set (Claude-4-Sonnet)

47.4%

best on the harder multi-document set (DeepSeek-R1)

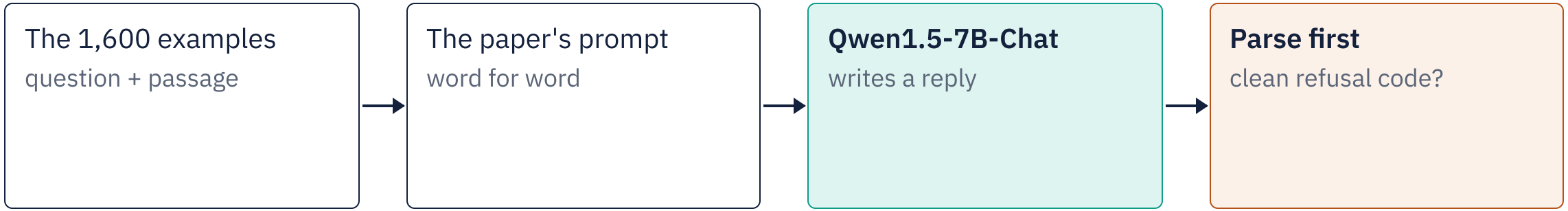
None

of the models tested reaches 80% on both answering and refusing

- **Detecting** that a refusal is needed is easier than saying **why**.
- Qwen models: answer accuracy jumps from 13.0% to 56.1% between 4B and 7B, but refusal accuracy stays below 17% at every size.
- Our target, **Qwen1.5-7B-Chat**: answer accuracy 56.1%, refusal accuracy about 5% (read from a small plot).

Our baseline · 1 of 4 · the scenario

A small open model answering from one passage, on free hardware



Yes: scored by code
Compare the code with the expected one. Free, instant, no judge.

No: ask the Gemini judge
Free text goes to the judge: answer or refusal, which code, how good.

Then the same metrics as the paper. Only 34% of Qwen's replies are a clean code, so the judge handles most of them.

Our baseline · 2 of 4 · for readers who have not done this

How a 7B model runs on a free GPU

1 **Google Colab.** A free notebook in the browser that lends a GPU for a few hours: a T4 with 15 GB of memory.

2 **4-bit loading.** The model needs about 15 GB at full size. Storing each weight in 4 bits brings it to about 5 GB, so it fits.

3 **One notebook.** It downloads our code from GitHub, runs, and saves every batch to Google Drive. If Colab disconnects, run it again and it resumes.

4 **Cost.** $0. The whole project stays far below the course limit of $20.

Our baseline · 3 of 4 · every change, written down

What we kept and what we changed

	Paper	Ours
Model and data	Qwen1.5-7B-Chat, 1,600 examples	The same **kept**
Prompt and sampling	Their prompt, temperature 1.0	The same **kept**
Hardware	4 A100 GPUs, 80 GB each	One free T4 **changed**
Precision and software	Likely full precision, vLLM	4-bit, Hugging Face Transformers **changed**
Judge	Claude-4-Sonnet	Gemini 3.5 Flash-Lite **changed**
What the judge grades	Every reply	Only replies without a clean code **changed**

Why each change: the free GPU and judge keep the cost at $0. Because of them our numbers are close in spirit to the paper's, not identical.

Our baseline · 4 of 4 · the numbers

Same story as the paper: the reason is almost never right

Metric	Paper	Ours
Answer accuracy	0.561	**0.690**
Refusal accuracy (right reason)	about 0.05	**0.114**
Answerable ones refused	not given	**0.165**
Needed refusals missed	not given	**0.456**

Honest status. 1,560 of 1,600 replies generated (40 ran out of GPU memory). **770 are graded so far**, limited by the free judge quota.

The baseline misses most refusals for **ambiguity (0.74)** and **contradiction (0.71)**.

Our answer accuracy is higher than the paper's, but the two are not directly comparable: different judge, precision and software.

The limitation: the baseline misses 46% of the refusals it should make

Idea: if the model first names what is wrong with the evidence, its decision should improve. Three versions of the prompt, all on the same 800 examples:

BASELINE

Answer or refuse

The paper's prompt: answer, or reply with a refusal code.

CONTROL

Add a final line

Same prompt, plus "end with a `FINAL:` line". Tests whether tidier output alone helps.

OUR IDEA

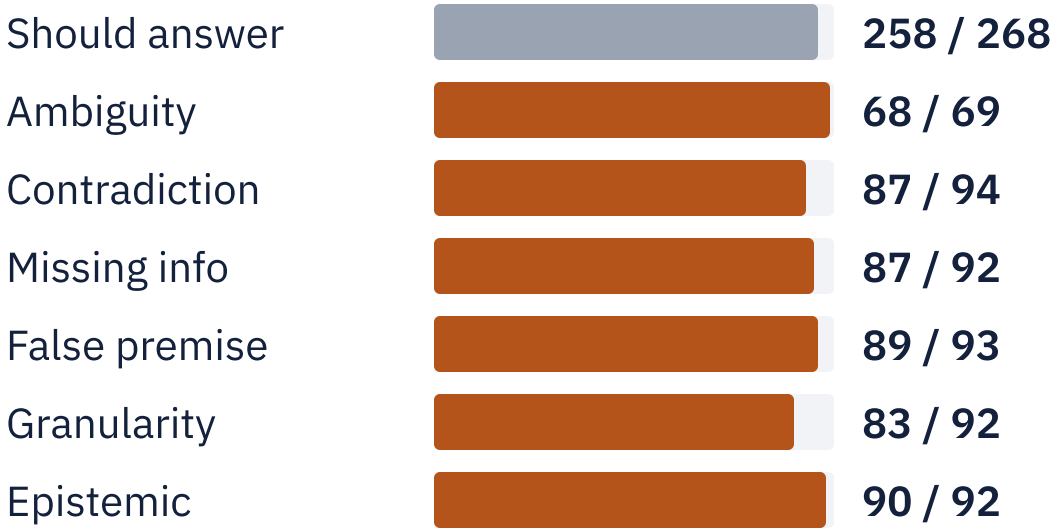
Diagnose first

First write `EVIDENCE_STATE:` (clear, or one of six problems), then `FINAL:` .

Extension 1 · 2 of 2 · the result

Asking the model to diagnose first made it worse

How often it wrote "clear"

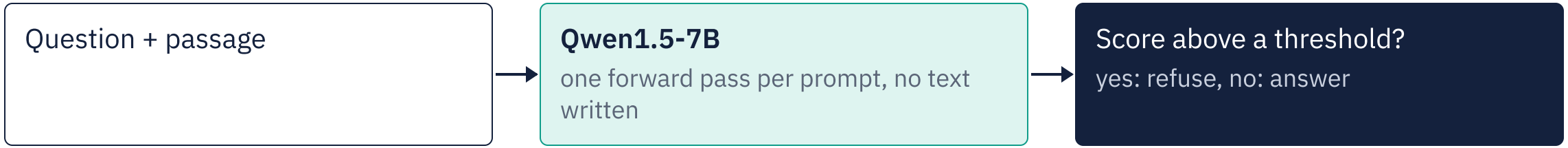


	Baseline	Control	Diagnose
Refusals caught	43.3%	35.8%	**1.9%**
Right reason	6.8%	3.3%	**0.0%**

762 of 800 times (95%) the model wrote "clear", even when it had to refuse. We report the result; we did not test why.

Extension 2 · 1 of 4 · the idea

Do not ask the model. Read it.



S1 · REFUSAL PROBABILITY

How likely is it that the reply **starts** with a refusal code?

S2 · YES OR NO

Ask "can this be answered from the passage?". Compare the chance of No with the chance of Yes.

S3 · LINEAR PROBE

A small classifier trained on the model's **hidden state**: the numbers inside it.

About 0.3 seconds per example on a T4. No judge, no API key, $0.

Variants of one question never sit on both sides of the test

100 source questions are split into **5 groups**. The threshold and the probe are tuned on 4 groups and tested on the fifth, five times.

tune	tune	tune	tune	**test**
tune	tune	**test**	tune	tune

...and so on for all five groups. Every example is scored by a rule that never saw its own question.

Why not a random split?

Variants of the same question would land in both parts, and the result would look better than it is.

Uncertainty

Intervals come from resampling whole questions, 2,000 times.

Rule fixed in advance

A rule is "better" only if its whole interval is above zero.

The signal is real, but it does not clearly beat the baseline

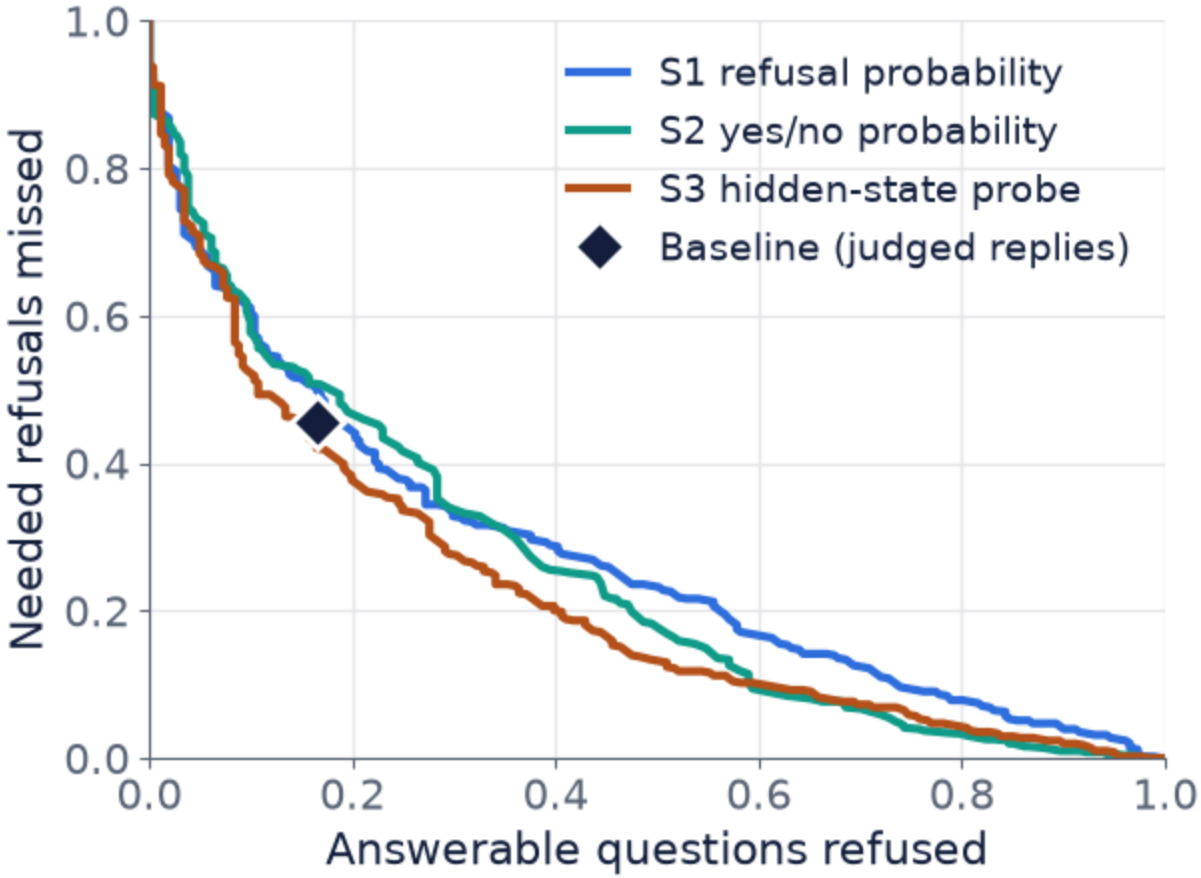
Readout	Signal	Decision	Difference from baseline	Verdict
Baseline	none	**0.690**	reference	
S1 refusal probability	0.726	0.658	-0.031 [-0.057, -0.005]	**slightly worse**
S2 yes or no	0.751	0.669	-0.020 [-0.047, +0.007]	unclear
S3 hidden-state probe	**0.797**	**0.718**	+0.028 [-0.014, +0.065]	unclear

Signal = AUROC: 0.5 is a coin flip, 1.0 is perfect. **Decision** = balanced accuracy on the 770 graded examples: 0.5 is chance. Brackets: 95% interval.

Read this carefully. The probe's +0.028 has an interval that includes zero. We do not claim an improvement.

Extension 2 · 3 of 4 · the same result as a picture

The baseline sits about where a simple threshold could put it



How to read it. Each line is one readout. Moving along a line trades one error for the other: refuse more, and you miss fewer needed refusals but refuse more answerable questions.

Lower left is better. The diamond is the baseline.

The diamond lies close to the S1 and S2 lines and just above the probe's. This is a visual observation, not a statistical test.

Extension 2 · 4 of 4 · where each readout helps

The probe finds more contradictions, and pays with false alarms

Failure type	Baseline	S1	S2	S3 probe
Ambiguity	.74 / .09	.66 / .11	.55 / .21	.45 / .23
Contradiction	.71 / .17	.63 / .24	**.23 / .50**	**.19 / .29**
Missing information	.56 / .22	.51 / .39	.49 / .34	.53 / .37
Epistemic mismatch	.30 / .12	.19 / .14	.30 / .19	.12 / .19
False premise	.19 / .33	.20 / .50	.16 / .47	.25 / .23
Granularity mismatch	.25 / .10	.19 / .23	.25 / .33	.19 / .35

Each cell: **needed refusals missed / answerable questions refused**, lower is better. Same 770 examples, 40 to 93 per cell, so read these as descriptive.

Conclusion

What we can say, what we cannot, and what a full run needs

We can say

- A small model's scores and hidden states carry real information about when to refuse (AUROC 0.73 to 0.80).
- Asking it to diagnose first backfires.

We cannot say

- That our method beats the baseline.
- Why the model answers "clear".
- That this holds for other models.

A full run needs

- All 1,560 replies graded.
- The 30-reply check of the judge.
- More seeds, models and benchmarks.

Central finding: a gap between **decodable** and **deployable**. The signal is there, but simple decision rules do not turn it into a clear gain.

References

- Muhamed et al. **RefusalBench.** EACL 2026, arXiv 2510.10390.
- Bai et al. **Qwen Technical Report.** arXiv 2309.16609, 2023.
- Kwiatkowski et al. **Natural Questions.** TACL 7, 2019.
- Kadavath et al. **Language Models (Mostly) Know What They Know.** arXiv 2207.05221.
- Burns et al. **Discovering Latent Knowledge.** ICLR 2023.
- Azaria and Mitchell. **The Internal State of an LLM Knows When It's Lying.** arXiv 2304.13734.
- Alain and Bengio. **Linear Classifier Probes.** arXiv 1610.01644.
- Yin et al. **Do LLMs Know What They Don't Know?** Findings of ACL 2023.
- Zhang et al. **R-Tuning.** NAACL 2024.
- Asai et al. **Self-RAG.** arXiv 2310.11511, 2023.
- Geifman and El-Yaniv. **Selective Classification for Deep Neural Networks.** 2017.
- Zheng et al. **Judging LLM-as-a-Judge.** NeurIPS 2023.
- Cameron, Gelbach, Miller. **Bootstrap with clustered errors.** REStat 2008.
- **Code, data, report:** github.com/aidepartmentrah-pixel/refusalbench-reproduction